

EXPERIMENT 18. Simple FOPC Parser

Aim: To implement a simple First-Order Predicate Calculus (FOPC) parser for basic logical expressions using Python.

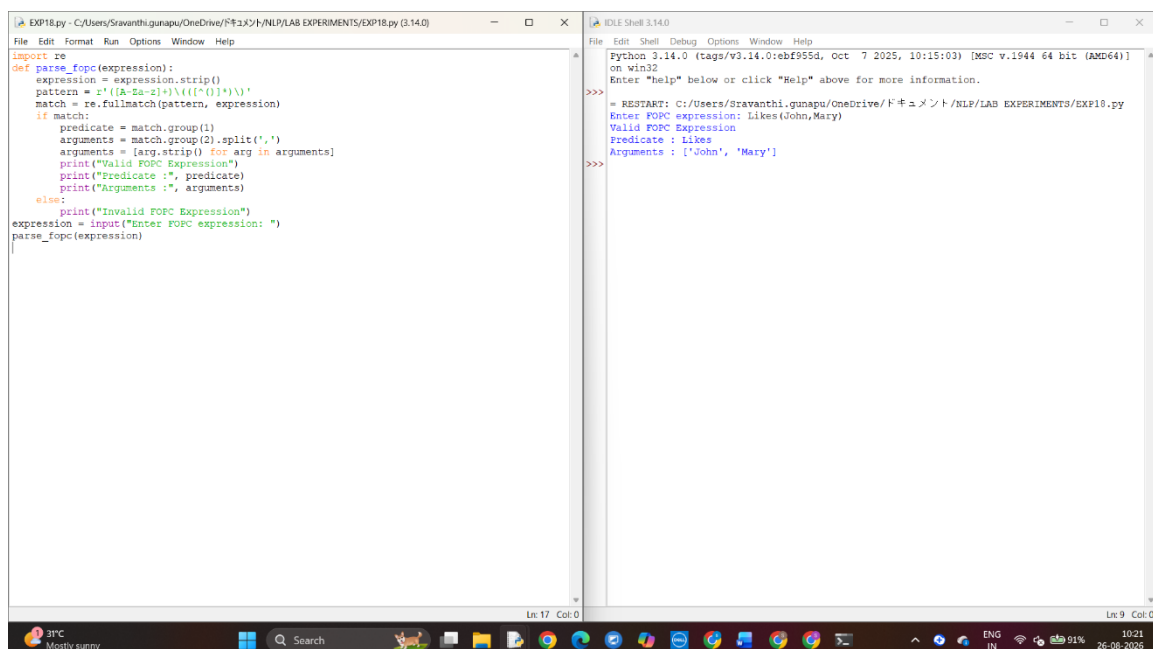
Program:

```
import re

def parse_fopc(expression):
    expression = expression.strip()
    pattern = r'([A-Za-z]+\s*)((\[^\s\)]*\s*)'
    match = re.fullmatch(pattern, expression)
    if match:
        predicate = match.group(1)
        arguments = match.group(2).split(',')
        arguments = [arg.strip() for arg in arguments]
        print("Valid FOPC Expression")
        print("Predicate :", predicate)
        print("Arguments :", arguments)
    else:
        print("Invalid FOPC Expression")

expression = input("Enter FOPC expression: ")
parse_fopc(expression)
```

Output:



The screenshot displays a Python IDE with two windows. The left window, titled 'EXP18.py', contains the source code for the FOPC parser. The code defines a function 'parse_fopc' that uses regular expressions to validate and parse logical expressions. It prompts the user to 'Enter FOPC expression: ' and then prints the results. The right window, titled 'IDLE Shell 3.14.0', shows the execution output. It displays the Python version (3.14.0), the file path, and the results of the parser for the input 'Likes(John,Mary)'. The output confirms it is a 'Valid FOPC Expression' and shows the predicate as 'Likes' and the arguments as 'John' and 'Mary'.

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>> = RESTART: C:/Users/Sravanthi.gunapu/OneDrive/ドキュメント/NLP/LAB EXPERIMENTS/EXP18.py
Enter FOPC expression: Likes(John,Mary)
Valid FOPC Expression
Predicate : Likes
Arguments : ['John', 'Mary']
>>>
```